

Experiment – 2

Interfacing of LEDs with Arduino Uno

Date:

Aim of the Experiment:

Interfacing of LEDs with Arduino Uno.

Objective:

- 1) Create an Arduino Sketch to control the intensity of one or more LEDs.
- 2) SOS LED Signaling using Arduino
- 3) Design a Knight Rider using Arduino

Components/Equipment Required:

Sl No.	Name of the Component/Equipment	Specification	Quantity
1	Arduino UNO R3	16MHz	1
2	Arduino UNO cable	USB Type A to B	1
3	Resistors (carbon type)	¼ watt (330Ω) or 220 Ω	4
4	LED	Red, Blue, Green, White	As per the requirement
5	Breadboard	840 Tie points	1
6	Jumper Wire	-----	As per the requirement

Objective 1

Circuit/Schematic Diagram:

Code:

Observation:

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Objective 2

SOS LED Signalling using Arduino

Problem Statement:

Design and implement an **SOS LED Signaling System** using an **Arduino Uno**, one external LED, and a **current-limiting resistor**. The system should transmit the internationally recognized **SOS distress signal** in Morse code by controlling the LED with different **ON and OFF durations**.

The Morse code for SOS is represented as:

S → ••• (Three short flashes)

O → — — — (Three long flashes)

S → ••• (Three short flashes)

Requirements

1. Connect an external LED to **Digital Pin 13** of the Arduino through a **220 Ω or 330 Ω current-limiting resistor**.
2. Configure Pin 13 as an **OUTPUT** in the setup() function.
3. In the loop() function, generate the SOS signal using the following timing:
 - **Dot (•):**
 - LED ON for 250 milliseconds
 - LED OFF for 250 milliseconds
 - **Dash (—):**
 - LED ON for 500 milliseconds
 - LED OFF for 500 milliseconds

Expected Output

The LED should continuously blink in the following pattern:

••• — •••

where:

- **Short blink (Dot)** = 250 ms ON, 250 ms OFF
- **Long blink (Dash)** = 500 ms ON, 500 ms OFF
- **Pause between successive SOS signals** = 3000 ms

Deliverables:

- **Circuit diagram.**

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- **Arduino program (.ino file).**
- **Demonstration of the working SOS LED signaling system.**

Circuit/Schematic Diagram:

Code:

Observation:

Objective 3

Circuit/Schematic Diagram:

Code:

Observation:

Post Experiment Questionnaire:

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- 1) List out the different types of Arduino boards available in the market and give one specific application of each board.
- 2) Differentiate between pull-up and pull-down resistor.
- 3) When to use pull-up or pull-down resistors?
- 4) What are PWM pins in Arduino? How many PWM pins are present in Arduino UNO?
- 5) Can we give power supply to the Arduino Uno through USB connection and DC power barrel jack simultaneously? Justify your answer.
- 6) Which voltage regulator is present in Arduino module?

Name of the Student

Registration No

Branch, Section